

CHAPTER 1: Big Data

Big data: (big data basic, sources, and challenges)

"Big data is a terms Used to describe data that has massive Volume, Comes in a Variety of Structures, and is generated at high Velocity. This kind of data Poses challenges to the traditional RDBMS Systems for storing and processing data. Big data is a paving way for newer approaches and Storing data."

Big data Sources: following are the Sources of big data:

- Mobile Use of the internet
- Cloud computing
- collaboration
- IP-Based Communication
- Social Media
- Video Streaming and Media distribution

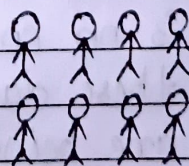
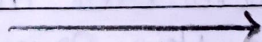
NoSQL - Language

RDBMS - concepts

MongoDB - Software

The Data Model:

- Old model: few Companies are generating data and all Others are consuming data.

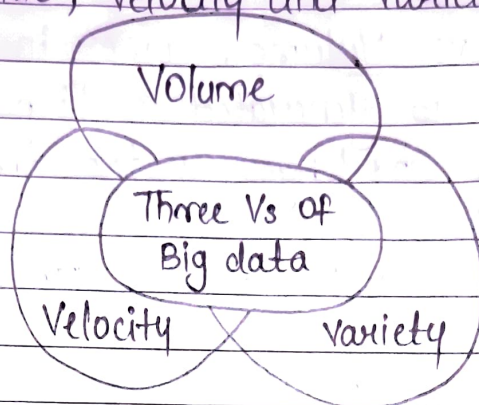


(Limited Source, more consumers)

- New Model: All of us are generating data, and all of us are consuming data.
(Many Sources, many consumers)

Three Vs Of Big data :

- We have defined big data as data with three Vs : Volume, Velocity and Variety.



- (1) Volume : Volume in big data means, the Sizes of the data. Various factors contribute to the Sizes of big data, as businesses are becoming more transaction - Oriented, we see ever increasing numbers of transactions; more devices are getting connected to the Internet, which is adding to the Volume.
- (2) Variety : The data generated from Various devices and Sources follows no fixed format or Structure. compared to text, CSV or RDBMS data Varies from text files, log files, Streaming Videos, photos, meter readings, Stock ticker data, PDFs audio and Various Other Unstructured formats.
There is no control Over the Structure of the data these days. New Sources and Structures of data are beings created at a rapid pace. so the Onus is On technology to find a Solution to analyze and visualize that huge Variety of data that is Out there.

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(3) Velocity: Velocity in big data is the Speed at which data is created and the Speed at which it's required to be processed. If data cannot be processed at the required Speed. It loses its Significance. Due to data Streaming in from social media sites Sensors, tickers, monitoring. It's important for the Organization to Speedily process data both when it's on move and when it's Static.

- Volume: Data at rest.
- Velocity: Data in motion.
- Variety: Data in many forms.

Usage of Big data:

- (1) Visibility: Accessibility to data in a timely fashion to relevant Stakeholders generates a tremendous amount of Value.
- (2) Discover and Analyze Information: Most of the Value of big data comes from when the data collected from outside Sources can be merged with the Organization's Internal data.
- (3) Segmentation and customizations: Big data enables Organizations to create tailor-made products and Services to meet Specific Segment needs. This can also be Used to the Social Sector to accurately Segment populations and target benefits Schemes for Specific needs.

(4) Aiding Decision Making:

Big Data can substantially minimize risks, improve decisions making, and Uncover available valuable insights (knowledges).

(5) Innovation:

Big data enables innovation of new ideas in the form of products and Services. It enables innovation in the existing Ones in Order to reach out to large Segments of people.

Big data Challenges:

(1) Policies and Procedures:

As more and more data is gathered, digitized, and moved around the globe, the policy and compliance issues becomes increasingly important. Data privacy, Security, intellectual property, and protection are of immense important to Organizations.

(2) Access to Data:

Accessing data for consumption is a challenges for big data projects. Some of the data may be available to third parties, and gaining access can be a legal, contractual challenges.

(3) Technology and Techniques:

New tools and technologies built Specifically to an address the needs of big data must be leveraged, rather than trying to address the aforementioned issues through legacy Systems.

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Legacy Systems and Big Data:

(1) Structure of Big data:

- Legacy Systems are designed to work with structured data where tables with columns are defined. The format of the data held in the columns is also known.
- Since big data can be unstructured, legacy systems created to perform fast queries and analysis through techniques like indexing based on particular data types held in various columns cannot be used to hold or process big data.

(2) Data Storage:

- Legacy Systems use big Servers and NAS and SAN Systems to store the data.
- Traditional legacy Systems typically work in a Scaleup model where more and more compute, memory, and storage needs to be added to a Server to meet the increased data needs.
- Hence, the processing time increases exponentially, which defeats the other important requirements of big data, which is a Velocity.

(3) Data Processing:

- The algorithms in Legacy Systems are designed to work with structured data such as strings and integers.
- They are also limited by the size of data.
- Thus, legacy Systems are not capable of handling the

Processing of Understand unstructured data, huge volumes of such data, and the speed at which the processing needs to be performed.

- We need new techniques for analyzing the data.

Big Data Technologies:

- The recent (recent) technology advancements that enables Organizations to make the most of its big data are the following:

1. New Storage and Processing technologies designed specifically for large Unstructured data.
2. Parallel Processing.
3. Clustering.
4. Large grid environments.
5. High connectivity and high throughput.
6. Cloud computing and Scale-out architectures.

CHAPTER 2: NoSQL

"NoSQL is a new way of designing Internet-Scale database solutions. It is not a product or technology but a term that defines a set of database technologies that are not based on the traditional RDBMS principles."

SQL: The idea of RDBMS was borne from E.F. Codd's 1970 whitepaper titled "A relational model of data for large shared data banks." The language used to query RDBMS systems is SQL (Sequel Query Language).
- The RDBMS systems are based on the concept of ACID transactions. ACID stands for Atomic, Consistent, Isolated, and Durable, where

- (1) Atomic: It implies either all changes of a transaction are applied completely or not applied at all.
- (2) Consistent: It means the data is in a consistent state after the transaction is applied. This means after a transaction is committed, the queries fetching a particular data will see the same result.
- (3) Isolated: It means the transactions that are applied to the same set of data are independent of each other. Thus, one transaction will not interfere with another transaction.
- (4) Durable: It means the changes are permanent in the systems and will not be lost in case of any failures.

NoSQL: NoSQL is a term used to non-relational databases.

(1) Definition:

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NoSQL doesn't have a formal definition. It represents a form of persistence / data storage mechanism that is fundamentally different from RDBMS, But if pushed to define NoSQL, here it is: NoSQL is an Umbrella term for data stores that don't follow the RDBMS principles.

(2) ACID Vs. BASE:

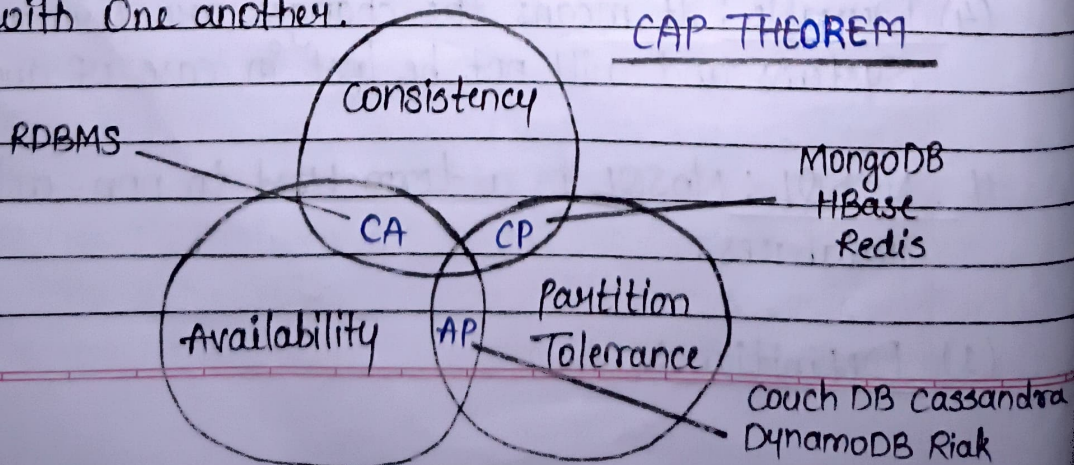
a) CAP Theorem (Brewer's Theorem):

- The ERIC BREWER Outlined the CAP theorem in 2000. The theorem states that when designing an application in a distributed environment there are three basic requirements that exist, namely consistency, availability, and partition tolerance.

(i) Consistency: It means that the data remains consistent after any operation is performed that changes the data, and that all users or clients accessing the application see the same updated data.

(ii) Availability: It means that the system is always available.

(iii) Partition tolerance: It means that the system will continue to function even if it is partitioned into groups of servers that are not able to communicate with one another.



b) The BASE:

- Eric Brewer coined the BASE acronym. BASE can be explained as,

- (i) Basically Available: It means the Systems will be available in terms of the CAP Theorem.
- (ii) Soft State: It indicates that even if no input is provided to the System, the state will change over time. This is in accordance to eventual consistency.
- (iii) Eventual consistency: It means the system will attain consistency in the long run, provided no input is sent to the system during that time.

- Hence BASE is in contrast with the RDBMS ACID transactions.

NoSQL Advantages and disadvantages:

a) Advantages of NoSQL:

- High Scalability: This scaling up approach fails when the transactions rates and fast response requirements increases. In contrast to this, the new generation of NoSQL databases is designed to scale out.
- Manageability and administration: NoSQL databases are designed to mostly work with automated repairs, distributed data, and simpler data models, leading to how low manageability and administration.

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- Low Cost: NoSQL databases are typically designed to work with a cluster of cheap commodity Servers, enabling the users to store and process more data at a low cost.
 - Flexible data Models: NoSQL databases have a very flexible data model, enabling them to work with any type of data; they don't comply with the rigid RDBMS data models. As a result, any application changes that involve updating the database Schema can be easily implemented.

b) Disadvantages of NoSQL:

- Maturity: Most NoSQL databases are pre-production versions with key features that are still to be implemented. Thus, when deciding on a NoSQL database, you should analyze the product properly to ensure the features are fully implemented and not still on the To-do list.
- Support: Support is one limitation that you need to consider. Most NoSQL databases are from start-ups which were open sourced. As a result, support is very minimal as compared to the enterprise software companies and may not have global reach or support resources.
- Limited Query Capabilities: Since NoSQL databases are generally developed to meet the scaling requirements of the web-scale applications, they provide limited querying capabilities. A simple querying requirement may

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involve Significant programming expertise.

- Administration: Although NoSQL is designed to provide a no-admin solution, it still required skill and effort for installing and maintaining the solution.
- Expertise: Since NoSQL is an evolution (evolving area), expertise on the technology is limited in the developer and administration community.

IMP # SQL Vs. NoSQL Databases:

IMP. Difference between SQL and NoSQL Databases.

Categories of NoSQL Databases:

<u>Category</u>	<u>Brief Description</u>	<u>for Eg.</u>
(1) Document - based	Data is stored in form of documents. for instance, { Name: "TestUser", Address: "Address1", Age: 8 }	MongoDB
(2) XML database	XML is used for storing data.	MarkLogic
(3) Graph databases	Data is stored as node collections. The nodes are connected via edges. A node is comparable to an object in a programming language.	GraphDB

(4) key-value store: Stores data as key-value pairs.

Eg.
Cassandra,
Redis,
memcached

Chapter 3: Introduction MongoDB (software)

"MongoDB is One of the leading NoSQL documents Store databases. It enables Organizations to handles and gain meaningful insights for from Bigdata."

MongoDB Design philosophy:

(1) Speed, Scalability, and Agility:

- The design team's goal when designing MongoDB was to Create a databases that was -first fast, massively Scalable, and easy to use. To achieve Speed and horizontal Scalability in a partitioned databases, as explained in the CAP Theorem, the consistency and transactional Support have to compromised. Thus, per this theorem, MongoDB provides high availability, Scalability, and partitioning at the cost of consistency and transactional Supports.

(2) Non-Relational Approach:

- MongoDB stores its data in BSON (BinaryJSON) documents where all the related data is placed together, which means everything is in One place.

Performance Vs. Features :

- In order to make MongoDB high performance and fast, a certain features commonly available in RDBMS Systems are not available in MongoDB. MongoDB is a document-Oriented DBMS where data is stored as documents. It doesn't support JOINS, and it doesn't have fully generalized transactions.
- However, it does provide support for Secondary indexes, it enables users to query using query documents, and it provides support for atomic updates at a per documents level. It provides a replica set, which is a form of master-slave replication with automated failover, and it has built-in horizontal scaling.

Running the Database Anywhere :

- One of the main design decisions was the ability to run the database from anywhere, which means it should be able to run on servers. The language used for implementing MongoDB is C++, python, PHP, jQuery and etc.

IMP # SQL Comparison : The following are the ways in which MongoDB is different from SQL.

JSON - Based Document Store :

- MongoDB uses a JSON-based (JavaScript Object Notation) documents store for the data. JSON / BSON offers a schema less model, which provides flexibility in terms of database design.